

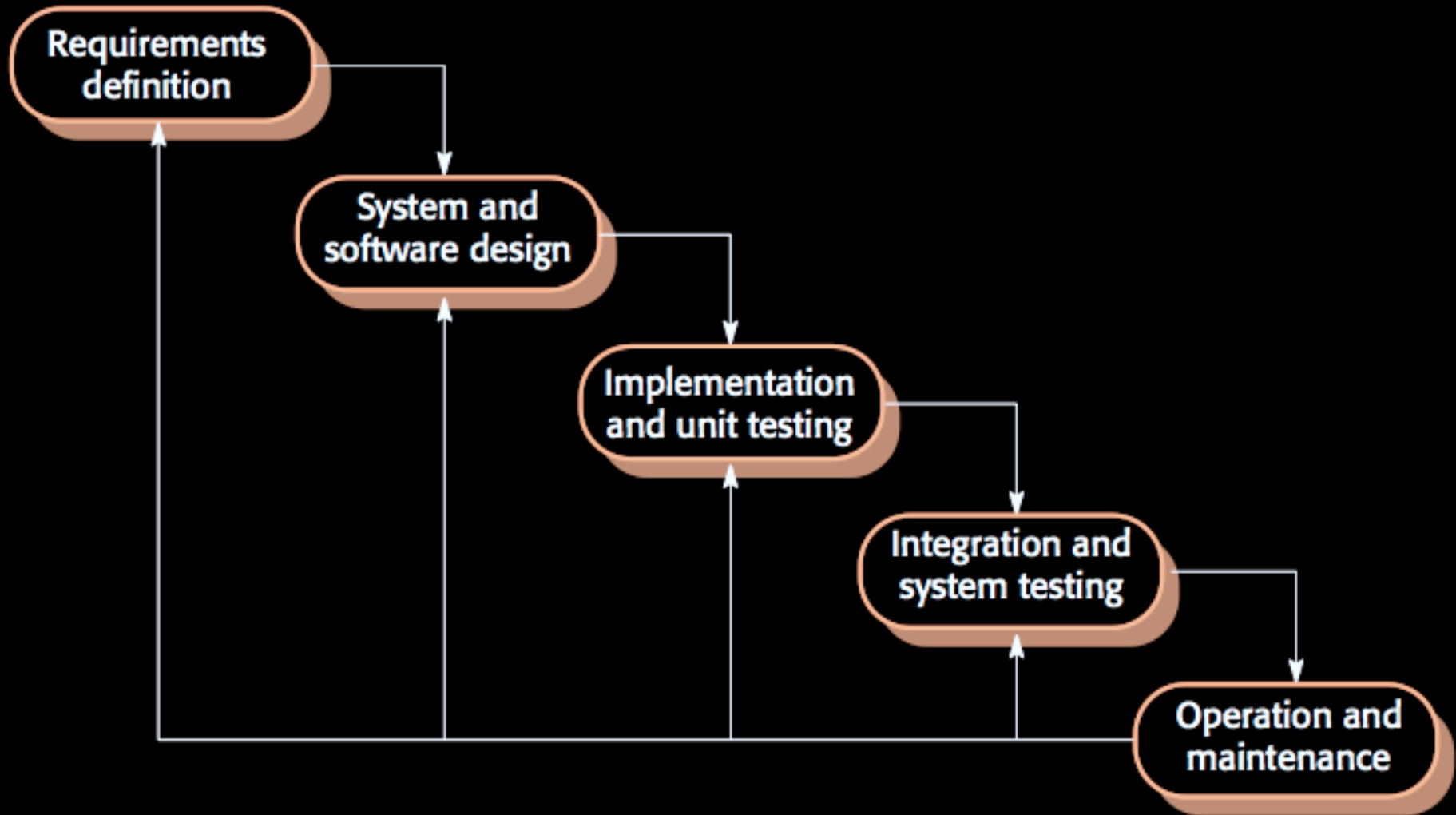
“THE” Software Development Process

- No such thing as "THE" software dev process !
- Different domains rely upon different processes
- Different companies have own in-house standards
- Similar concepts can have different names
- Different experts advocate their own approach
- All will try and convince you that theirs is the best !

Standardisation

- There is limited agreement/standardisation
- We'll try to put across something coherent
- Present some key concepts and processes
- Might differ from what your client advocates
- Might differ from what you do when you graduate
- But majority of the fundamentals you will recognise

Waterfall Model



Realism

- People dis waterfall model as naïve and unrealistic
 - But remember - it's a simplification & abstraction
 - A tool to help explain activities of SW development
 - Useful for undergrad Computer Science courses !
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- It's not *exactly* what happens in the real world,
 - But it is a good start in trying to understand it !

Waterfall versus Agile

- You've probably heard of Agile Development
- Advocates of Agile often denigrate Waterfall
- "It's old fashioned & unrealistic" [see previous slide]
- Agile methods undertake all stages in parallel
- Using short-burst "sprints" to deliver functionality
- All stages at the same time ?
- So you are eliciting requirements with one hand and writing code with the other ?

Key Development Activities

Requirements

- What should the system do ?
- What are the needs of the users ?
- What are the needs of the host organisation ?
- What are interoperability needs of existing systems

We need to think about functional elements
(stuff that system should do)

And also non-functional elements

(quality properties of system operation)

Security, Reliability, Ease of Use, Maintainability etc.

Design

- Before we implement anything, we need an overall structure for the system
- What objects, architectures, files, databases, servers, services etc. should we create ?
- Only then do we know the places where requirements should be implemented
- Also helps divide up work between team members

Implementation

You are familiar with programming, but there is more to implementation than just “cutting code”...

- Parallel working and code sharing
- API partitioning and “firewalling”
- Versioning, integration and config management
- Development environments and automated build
- Automated testing

Docs and training material also need writing !

Verification & Validation

- Demonstrates that system meets needs of client
- Involves checking, reviewing, evaluating & testing
- Testing is the most common V & V activity
- So you'll be doing plenty of that !
- User evaluation is the most undervalued
- You'll come up with excuses why you didn't do it
;o)

What next ?

In later weeks we'll consider all stages in detail

Remember: waterfall model is a big simplification

But it's nice way of structuring content of this unit :o)